

Project Design Phase Proposed Solution Template

Date	19 July 2026
Team ID	
Project Name	Nightjar & Co. – MERN BookStore
Maximum Marks	2 Marks

Project Design Phase – Proposed Solution Template

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Readers and students often struggle to find and purchase books through a simple, secure, and well-organized online platform. Many existing websites have complicated navigation, limited search capabilities, and lengthy purchasing processes.
2	Idea / Solution Description	Nightjar & Co. – MERN BookStore is a full-stack web application developed using the MERN stack (MongoDB, Express.js, React, and Node.js). It enables users to register securely, browse books by category, search for books, view book details, manage a shopping cart, and place orders through an intuitive interface.
3	Novelty / Uniqueness	The application combines secure JWT-based authentication, responsive design, efficient book search and filtering, category-based browsing, and a clean user interface to provide a smooth online book shopping experience. Its modular MERN architecture also makes it easy to maintain and extend.
4	Social Impact / Customer Satisfaction	The platform improves access to books for students, professionals, and book enthusiasts by allowing them to browse and purchase books from anywhere. A user-friendly interface and secure authentication enhance customer satisfaction and encourage digital adoption.
5	Business Model (Revenue Model)	Revenue can be generated through direct book sales, premium memberships, featured book promotions, advertising partnerships with publishers, and affiliate marketing for selected books or educational resources.
6	Scalability of the Solution	The application can be scaled by integrating online payment gateways, order tracking, personalized recommendations, book reviews, wishlists, e-books, multi-vendor support, and cloud deployment to support a growing number of users and products.